Residual-Shape Calibration and Empirical Drift Construction: Companion Note

Abstract

This note accompanies *Optimal Sizing of Leveraged Crypto Long–Short Positions under Kou Jump-Diffusion via Killed Moments*. It documents the empirical construction used in the paper's showcase. The construction begins from the main paper's decomposition of log-price changes into an expected log-return component and a zero-mean Kou residual process. A normalized, causal exponentially weighted mean (EWM) estimates the conditional location of the observed log-price increments, while a penalized empirical-characteristic-function (ECF) criterion estimates only the diffusion, jump, and dependence shape of the resulting residual increments. The forward expected log-return mean is then supplied from the endpoint EWM forecast plus role-specific carry, or from a portfolio manager's view. Finally, the fitted shape and supplied mean are combined to obtain the log-SDE coefficient and the expected-price growth exponent used by the empirical analysis.

1. Location–shape decomposition and empirical workflow

Let $S_{i,t_n} > 0$ be the observed price of asset i at equally spaced dates $t_n = n\Delta t$, let $X^{\mathrm{obs}}_{i,t_n} = \log(S_{i,t_n}/S_{i,0})$, and let $\Delta X^{\mathrm{obs}}_{i,t_n} := X^{\mathrm{obs}}_{i,t_n} - X^{\mathrm{obs}}_{i,t_{n-1}} = \log(S_{i,t_n}/S_{i,t_{n-1}})$ be the observed log-price increment. Here ΔX denotes a discrete increment, whereas $\mathrm{d}X$ in the main text denotes a continuous-time differential. The superscript "obs" distinguishes the observed sample path from the forward process $X_{i,t}$ in the main paper. At the population level, the observed increment is decomposed as

$$\Delta X^{\mathrm{obs}}_{i,t_n} = \mathbb{E}\big[\Delta X^{\mathrm{obs}}_{i,t_n} \mid \mathscr{F}_{t_{n-1}}\big] + \Delta X^{\mathrm{res}}_{i,t_n}, \qquad \mathbb{E}\big[\Delta X^{\mathrm{res}}_{i,t_n} \mid \mathscr{F}_{t_{n-1}}\big] = 0. \tag{1}$$

The first term is location; the law of $\Delta X^{\mathrm{res}}_{i,t_n}$ supplies shape. The conditional expectation $\mathbb{E}[\Delta X^{\mathrm{obs}}_{i,t_n} \mid \mathscr{F}_{t_{n-1}}]$ is an estimand, not a quantity intrinsically tied to a particular filter. Depending on the application and information set, it may be estimated using, for example, a rolling or kernel mean, a parametric predictive regression, a state-space filter, or an exponentially weighted mean. In this empirical study we use the normalized causal EWM defined in Section 3, with a one-month half-life matched to the analysis horizon. This choice specifies the historical location estimator; it does not alter the residual Kou law or make the EWM part of the ECF criterion. In the forward specification of the main paper, the corresponding continuous-time identity is

$$\mathrm{d}X_{i,t} = g_i\,\mathrm{d}t + \mathrm{d}X^{\mathrm{res}}_{i,t}, \tag{2}$$

where $g_i = \mathbb{E}[X_{i,t}]/t$ is the externally supplied expected log-return mean. Section 2 defines the residual Kou law, Section 3 estimates the historical conditional location by EWM, Section 4 fits the residual shape by ECF, and Section 5 constructs the parameter set used in the empirical analysis.

Stage	Input and operation	Estimated or fixed quantity	Quantity passed forward
1	Positive prices S_{i,t_n}	Observed increments $\Delta X^{\mathrm{obs}}_{i,t_n}$	Increment sample on the model time scale
2	Zero-mean residual Kou law	Location–shape decomposition	Model for $\Delta X^{\mathrm{res}}_{i,t_n}$
3	Chosen location estimator: causal EWM subtraction	$d^{\mathrm{EWM}}_{i,n}$ and residual increments	Sample supplied to ECF
4	Penalized ECF	Residual shape ϑ	$\sigma_i, \lambda_i, p_i, \eta_{i,+}, \eta_{i,-}, \rho$
5	Endpoint EWM, annual r_i^{role}, and fitted shape	g_i and derived drifts	Main-paper parameter set

Table 1: Model-to-estimator map. The ECF estimates only residual shape. The directional mean enters when the final main-paper parameter set is assembled.

2. Zero-mean residual Kou law

The residual component in (1) is modeled by the same Kou process used in the main text, normalized to have zero expected log drift:

$$\mathrm{d}X^{\mathrm{res}}_{i,t} = -\lambda_i \bar{J}_i \,\mathrm{d}t + \sigma_i \,\mathrm{d}B_{i,t} + \mathrm{d}\left(\sum_{n=1}^{N_{i,t}} J_i^{(n)}\right), \qquad \mathbb{E}[X^{\mathrm{res}}_{i,t}] = 0. \tag{3}$$

Here B_1 and B_2 are standard Brownian motions satisfying $\mathrm{d}\langle B_1, B_2\rangle_t = \rho\,\mathrm{d}t$. The Poisson processes N_1, N_2, with respective intensities λ_1, λ_2, are independent of each other and of (B_1, B_2). For each asset, the jumps $J_i^{(n)}$ are independent and identically distributed, and the two jump sequences are independent, with Kou density

$$f_i(j) = \frac{p_i}{\eta_{i,+}} e^{-j/\eta_{i,+}} \mathbf{1}_{\{j>0\}} + \frac{1-p_i}{\eta_{i,-}} e^{j/\eta_{i,-}} \mathbf{1}_{\{j<0\}}. \tag{4}$$

The mean log jump and price-jump compensator are

$$\bar{J}_i := \mathbb{E}[J_i] = p_i \eta_{i,+} - (1-p_i)\eta_{i,-}, \qquad \chi_i := \mathbb{E}[e^{J_i} - 1] = \frac{p_i}{1-\eta_{i,+}} + \frac{1-p_i}{1+\eta_{i,-}} - 1. \tag{5}$$

Thus $-\lambda_i \bar{J}_i$ exactly compensates the expected compound-Poisson log jump in (3). No $-\frac{1}{2}\sigma_i^2$ term is needed because this normalization concerns the expected *log* increment, and $\mathbb{E}[\mathrm{d}B_{i,t}] = 0$.

The residual law is determined by the eleven shape parameters

$$\vartheta = \big((\sigma_i, \lambda_i, p_i, \eta_{i,+}, \eta_{i,-})_{i=1,2}, \rho\big).$$

Here $\sigma_i > 0$ is diffusion volatility, $\lambda_i > 0$ is jump intensity, $p_i \in (0,1)$ is the upward-jump probability, $\eta_{i,+} > 0$ and $\eta_{i,-} > 0$ are the mean positive and absolute mean negative log-jump sizes, and $|\rho| < 1$ is Brownian correlation. The jump characteristic function is

$$\varphi_{J_i}(u) = \frac{p_i}{1 - iu\eta_{i,+}} + \frac{1-p_i}{1 + iu\eta_{i,-}}. \tag{6}$$

The price-jump compensator requires $\eta_{i,+} < 1$; moments through order K require $K\eta_{i,+} < 1$. The bivariate residual characteristic exponent is

$$\begin{aligned}\Psi^{\mathrm{res}}_{\vartheta}(u,v) = {} & i(-\lambda_1 \bar{J}_1 u - \lambda_2 \bar{J}_2 v) \\ & - \frac{1}{2}\big(\sigma_1^2 u^2 + \sigma_2^2 v^2 + 2\rho\sigma_1\sigma_2 uv\big) \\ & + \lambda_1\big(\varphi_{J_1}(u) - 1\big) + \lambda_2\big(\varphi_{J_2}(v) - 1\big).\end{aligned} \tag{7}$$

3. EWM estimation and residual-increment construction

The conditional mean in (1) is unknown. For the empirical horizon $T = 1/12$, set $\beta = 2^{-\Delta t/T}$ and estimate its annualized path by the normalized causal EWM

$$d_{i,n}^{\mathrm{EWM}} := \frac{1}{\Delta t} \frac{\sum_{j=1}^{n} \beta^{n-j} \Delta X_{i,t_j}^{\mathrm{obs}}}{\sum_{j=1}^{n} \beta^{n-j}}. \tag{8}$$

Normalizing the weights avoids dependence on an arbitrary recursive starting value. The residual increment subtracts the lagged EWM estimate directly from the observed increment:

$$\boxed{\Delta X_{i,t_n}^{\mathrm{res}} := \Delta X_{i,t_n}^{\mathrm{obs}} - d_{i,n-1}^{\mathrm{EWM}} \Delta t}, \qquad n = 2, \ldots, N. \tag{9}$$

The lag ensures that the current increment does not enter its own conditional-mean estimate. No further sample demeaning is applied. Thus $d_{i,n-1}^{\mathrm{EWM}} \Delta t$ is the sole estimated location removed before ECF calibration, and the exact reconstruction is

$$\boxed{\Delta X_{i,t_n}^{\mathrm{obs}} = d_{i,n-1}^{\mathrm{EWM}} \Delta t + \Delta X_{i,t_n}^{\mathrm{res}}}. \tag{10}$$

Under a correctly specified conditional-mean filter, the residual increment has zero conditional mean in the population. Its realized sample average need not equal zero and is retained as sampling variation and as a diagnostic of the EWM specification. Consequently, the zero-mean residual Kou law is a population restriction rather than a mechanically imposed sample identity.

The same operation is a multiplicative detrending of the positive price process. Define only the required cumulative removed location,

$$C_{i,n} := \sum_{k=2}^{n} d_{i,k-1}^{\mathrm{EWM}} \Delta t, \qquad C_{i,1} := 0, \tag{11}$$

and set

$$S_{i,t_n}^{\mathrm{res}} := S_{i,t_n} e^{-C_{i,n}}, \qquad X_{i,t_n}^{\mathrm{res}} := \log \frac{S_{i,t_n}^{\mathrm{res}}}{S_{i,0}}. \tag{12}$$

Then

$$\Delta X_{i,t_n}^{\mathrm{res}} = X_{i,t_n}^{\mathrm{res}} - X_{i,t_{n-1}}^{\mathrm{res}} = \log \frac{S_{i,t_n}^{\mathrm{res}}}{S_{i,t_{n-1}}^{\mathrm{res}}}, \qquad S_{i,t_n} = S_{i,t_n}^{\mathrm{res}} e^{C_{i,n}}. \tag{13}$$

Thus the return transformation is reversible. Arithmetic price demeaning is not used because it can produce negative values, depends on the price unit, and does not preserve the exponential-Lévy structure.

Figure 1 displays this decomposition in price space. The EWM curve is not an exponentially weighted average of price levels. It is the multiplicative location component reconstructed from the same lagged causal log-return means removed in (9): $\widehat{S}_{i,t_n}^{\mathrm{EWM}} := S_{i,0} \exp(C_{i,n})$. Consequently, deviations of the observed price from this curve are retained in S_{i,t_n}^{res} and passed to the residual-increment calibration rather than being smoothed away.

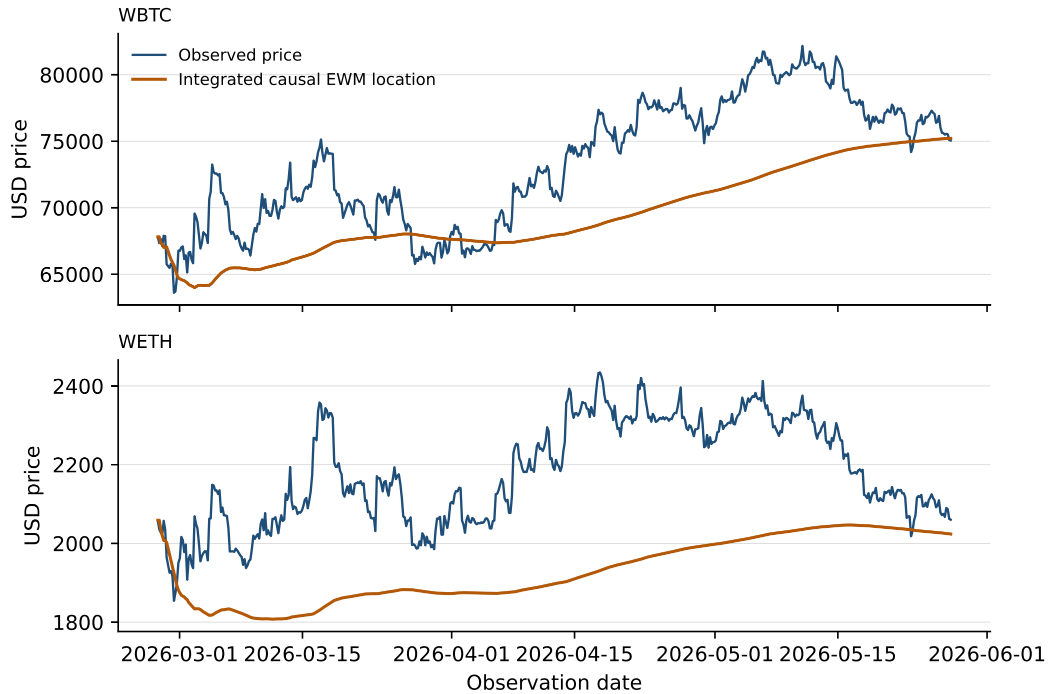

Figure 1: Observed WBTC and WETH prices and the price-level location components obtained by integrating the lagged causal EWM log-return means. The EWM half-life is $T = 1/12$ year. The orange curves are $S_{i,0}\exp(C_{i,n})$, not EWM smoothers applied directly to price levels.
Result-generation script: `jobs/calibrate_eth_btc.py`

The 540 observed WETH/WBTC increment pairs yield 539 residual pairs. Here $T/(\Delta t) = 182.5$, $\beta = 0.9962091$, and the equivalent pandas-style span is 526.584. The half-life-T choice is a transparent horizon-matching convention, not an estimated optimum (Cox, 1961; Tiao and Xu, 1993).

4. ECF estimation of residual shape

The residual characteristic exponent in (7) gives the model characteristic function over one observation interval:

$$\phi_{\vartheta}^{\mathrm{res}}(u,v;\Delta t) = \exp\big(\Delta t\,\Psi_{\vartheta}^{\mathrm{res}}(u,v)\big). \tag{14}$$

The residual sample contains $N_{\mathrm{res}} = N-1$ pairs $(\Delta X_{1,t_n}^{\mathrm{res}}, \Delta X_{2,t_n}^{\mathrm{res}})$, $n = 2,\ldots,N$. Its empirical characteristic function is

$$\widehat{\phi}_{N_{\mathrm{res}}}(u,v) = \frac{1}{N_{\mathrm{res}}}\sum_{n=2}^{N}\exp\big(i(u\,\Delta X_{1,t_n}^{\mathrm{res}} + v\,\Delta X_{2,t_n}^{\mathrm{res}})\big). \tag{15}$$

The normalized ECF distance over the frequency grid $\mathscr{G}$ is

$$Q_{N_{\mathrm{res}}}^{\mathrm{ECF}}(\vartheta) = \frac{\sum_{(u,v)\in\mathscr{G}} w(u,v)\left|\widehat{\phi}_{N_{\mathrm{res}}}(u,v) - \phi_{\vartheta}^{\mathrm{res}}(u,v;\Delta t)\right|^2}{\sum_{(u,v)\in\mathscr{G}} w(u,v)}. \tag{16}$$

Because a short high-frequency sample leaves a pronounced intensity–jump-size ridge, the reported fit uses transparent soft anchors:

$$
\begin{aligned}
\widehat{\vartheta} &= \arg\min_{\vartheta\in\mathscr{V}} \mathscr{Q}_{N_{\mathrm{res}}}(\vartheta), \\
\mathscr{Q}_{N_{\mathrm{res}}}(\vartheta) &= Q^{\mathrm{ECF}}_{N_{\mathrm{res}}}(\vartheta) + \omega_\lambda \sum_{i=1}^{2} (\log\lambda_i - \log\widetilde{\lambda}_i)^2 \\
&\quad + \omega_\eta \sum_{i=1}^{2} \sum_{s\in\{+,-\}} (\log\eta_{i,s} - \log\widetilde{\eta}_{i,s})^2.
\end{aligned} \tag{17}
$$

Here $\widetilde{\lambda}_i$ and $\widetilde{\eta}_{i,s}$ are the initializers described in Section 7. The application uses $(\omega_\lambda, \omega_\eta) = (0.05, 0.02)$; setting both to zero recovers pure ECF matching. Crucially, neither g_i, μ_i^X, nor μ_i is an ECF coordinate. The ECF literature motivates this shape estimation through known characteristic functions (Feuerverger and McDunnough, 1981; Singleton, 2001; Jiang and Knight, 2002); it does not by itself establish that the residual increments are stationary independent Kou increments.

5. Construction of the empirical parameter set

The ECF output $\widehat{\vartheta}$ supplies residual shape but no directional mean. The endpoint of the EWM path is the annualized forecast already used in the main text:

$$d_i^{\mathrm{EWM}} := d_{i,N}^{\mathrm{EWM}}. \tag{18}$$

The forward expected log-return mean is supplied as

$$g_i = \begin{cases} d_i^{\mathrm{EWM}} + r_i^{\mathrm{role}}, & \text{benchmark}, \\ d_i^{\mathrm{EWM}} + v_i + r_i^{\mathrm{role}}, & \text{portfolio-manager overlay } v_i, \\ d_i^{\mathrm{PM}} + r_i^{\mathrm{role}}, & \text{absolute portfolio-manager forecast}. \end{cases} \tag{19}$$

Here r_i^{role} is the lending rate for the long asset and the borrowing rate for the short asset. Every term in (19) is annualized. For an ordered long–short assignment (L,S), the benchmark is

$$g_L(L,S) = d_L^{\mathrm{EWM}} + r_{\mathrm{supply},L}, \qquad g_S(L,S) = d_S^{\mathrm{EWM}} + r_{\mathrm{borrow},S}. \tag{20}$$

The borrowing rate enters positively because it is the growth rate of the short debt liability. The selected orientation maximizes $g_L(L,S) - g_S(L,S)$.

No additional sample-mean adjustment enters g_i: the EWM estimate, financing, and any manager view determine the forward mean. With this g_i, (2) becomes the main-text log-price model

$$\mathrm{d}X_{i,t} = \mu_i^X\,\mathrm{d}t + \sigma_i\,\mathrm{d}B_{i,t} + \mathrm{d}\left(\sum_{n=1}^{N_{i,t}} J_i^{(n)}\right), \qquad \mu_i^X = g_i - \lambda_i\bar{J}_i. \tag{21}$$

Indeed, $\mathbb{E}[X_{i,t}]/t = \mu_i^X + \lambda_i\bar{J}_i = g_i$. Converting from expected log return to the expected-price growth convention of the main paper gives

$$\boxed{\mu_i^X = g_i - \lambda_i\bar{J}_i}, \qquad \boxed{\mu_i = g_i + \frac{1}{2}\sigma_i^2 + \lambda_i(\chi_i - \bar{J}_i)}, \tag{22}$$

where $\mathbb{E}[S_{i,t}/S_{i,0}] = e^{\mu_i t}$. Equivalently, the residual shape contributes

$$\mu_i^{\text{res}} := \frac{1}{2}\sigma_i^2 + \lambda_i(\chi_i - \bar{J}_i), \qquad \mu_i = g_i + \mu_i^{\text{res}}. \tag{23}$$

The $\frac{1}{2}\sigma_i^2$ term appears only in the conversion to expected price growth, not in g_i or μ_i^X.

After calibration, the main text suppresses estimator hats. The parameter set passed to the moment and liquidation calculations is

$$((\mu_i, \sigma_i, \lambda_i, p_i, \eta_{i,+}, \eta_{i,-})_{i=1,2}, \rho). \tag{24}$$

Initial prices, the horizon, liquidation thresholds, and health-buffer choices are contract or evaluation inputs and are supplied separately; they are not outputs of the EWM–ECF calibration. Thus assembly is not merely concatenation: ECF supplies the residual shape, the external step supplies g_i, and (22) produces the μ_i required by the moment code. The associated log-SDE coefficient is μ_i^X. If a manager instead supplies an expected-price growth exponent μ_i^{view}, it must first be converted to $g_i = \mu_i^{\text{view}} - \mu_i^{\text{res}}$.

The liquidation-driving log-price spread remains $Z_t = X_{1,t} - X_{2,t}$. The limited information about expected returns in short samples motivates keeping directional drift outside the shape fit (Merton, 1980). A bootstrap for the full procedure should rerun EWM estimation, residual construction, ECF optimization, orientation, and parameter assembly.

6. Frequency grid

The shape vector ϑ has the eleven free components stated above; neither μ_1 nor μ_2 is an ECF coordinate. Residual-increment magnitudes depend on the sampling interval, so fixed raw Fourier frequencies are poorly portable. Let s_1, s_2, and s_Z be IQR-based robust scales, $s = \max\{(q_{0.75} - q_{0.25})/1.349, 10^{-8}\}$, for the sample vectors ΔX_1^{res}, ΔX_2^{res}, and $\Delta X_1^{\text{res}} - \Delta X_2^{\text{res}}$. Dimensionless base frequencies are mapped to raw frequencies by division by the corresponding scale. The grid contains the following groups, with every point accompanied by its conjugate $(-u, -v)$:

- marginal points $(a/s_1, 0)$ and $(0, a/s_2)$ for $a \in \mathscr{A}_{\text{marg}}$;
- same-sign and mixed-sign joint points $(a/s_1, b/s_2)$ and $(a/s_1, -b/s_2)$ for $a, b \in \mathscr{A}_{\text{joint}}$, which are informative about ρ;
- spread points $(a/s_Z, -a/s_Z)$ for $a \in \mathscr{A}_Z$, which directly match the log-ratio process governing liquidation; and
- high-base marginal and spread points for $a \in \mathscr{A}_{\text{tail}}$, included to retain information about jump-tail shape.

The dimensionless base nodes are

$$\begin{aligned}
\mathscr{A}_{\text{marg}} &= \{0.1, 0.2, 0.5, 0.75, 1, 1.5, 2, 3, 5, 7.5, 10\}, \\
\mathscr{A}_{\text{joint}} &= \{0.25, 0.5, 1, 2, 3, 5\}, \\
\mathscr{A}_Z &= \{0.1, 0.2, 0.5, 0.75, 1, 1.5, 2, 3, 5\}, \\
\mathscr{A}_{\text{tail}} &= \{8, 12, 16, 20, 25\}.
\end{aligned}$$

Points whose absolute raw coordinate exceeds 5,000 are discarded to avoid numerical aliasing. Including conjugate pairs, the unfiltered construction has 236 frequency evaluations; this cap leaves 234 for the application sample. Weights depend on dimensionless rather than raw frequency:

$$w(a, b) = \exp\{-0.02(a^2 + b^2)\}.$$

The ordinary spread group receives an additional factor of two. Thus the criterion emphasizes the barrier-driving ratio without making the choice of observation interval determine the effective frequency weights.

7. Reparameterisation and initialisation

Unconstrained reparameterisation. The parameter constraints stated above are enforced automatically via the change of variables

$$\sigma_i = e^{a_i}, \quad \lambda_i = e^{b_i}, \quad p_i = \mathrm{sigmoid}(c_i), \quad \eta_{i,+} = \eta_{i,+}^{\max}\mathrm{sigmoid}(d_i), \quad \eta_{i,-} = e^{e_i}, \quad \rho = \tanh(f), \tag{25}$$

where $\mathrm{sigmoid}(x) = (1+e^{-x})^{-1}$. The optimized vector is the eleven-dimensional shape coordinate

$$\tau_{\text{shape}} = (a_1, b_1, c_1, d_1, e_1, a_2, b_2, c_2, d_2, e_2, f).$$

There is no drift coordinate: at every objective evaluation, the residual price-growth correction μ_i^{res} for each asset is reconstructed from the candidate shape by (23). The optimization uses box bounds in the transformed space. For both assets the code uses a scaled sigmoid with

$$\eta_{i,+}^{\max} \le (1-\varepsilon)/K, \qquad i = 1, 2,$$

so that $K\eta_{i,+} < 1$ is enforced for $i = 1, 2$, ensuring finiteness of the exponential moments used through order K.

Threshold initialiser. The characteristic-function objective is non-linear and non-convex, so the optimisation is initialised by a threshold-based decomposition following the logic of Mancini (2009) for discretely observed jump-diffusions. Let $q_i = \mathrm{median}(\Delta X_{i,t_n}^{\text{res}})$ and let s_i be the IQR robust scale. For a frequency-aware threshold constant $c \in [3, 5]$, define candidate jump observations

$$\mathscr{J}_i = \{n : |\Delta X_{i,t_n}^{\text{res}} - q_i| > c\, s_i\}, \qquad \mathscr{C}_i = \{2, \dots, N\} \setminus \mathscr{J}_i.$$

Initial volatility and correlation values use the continuous subset $\mathscr{C}_i$, while the jump-size means are anchored by peak-over-threshold excess means and the intensities are corrected by a variance/fourth-cumulant estimate. Schematically:

$$\begin{aligned}
p_i^{(0)} &= \frac{|\mathscr{J}_{i,+}|}{|\mathscr{J}_i|} \quad \text{clamped to the admissible probability range,} \\
\eta_{i,+}^{(0)}, \eta_{i,-}^{(0)} &= \text{POT mean-excess estimates,} \\
\lambda_i^{(0)} &= \text{two-moment estimate from variance and fourth cumulant,} \\
\sigma_i^{(0)} &= \mathrm{sd}(\Delta X_{i,t_n}^{\text{res}} : n \in \mathscr{C}_i)/\sqrt{\Delta t}, \\
\rho^{(0)} &= \mathrm{corr}(\Delta X_1^{\text{res}}, \Delta X_2^{\text{res}})_{n \in \mathscr{C}_1 \cap \mathscr{C}_2}.
\end{aligned}$$

No sample-mean or drift initializer enters the optimization; zero expected log drift is imposed through (3), with the corresponding price-growth correction given by (23). The threshold filter is not the final estimator. It is a warm start that places the optimizer in a sensible region of parameter space; see the remark in Mancini (2009).

Multi-start cloud. Because $\mathscr{Q}_{N_{\mathrm{res}}}(\tau_{\mathrm{shape}})$ may have local minima, the optimizer is run from a structured cloud. It contains the unperturbed initializer and six deterministic intensity–size ridge starts with

$$\lambda_i \mapsto c\lambda_i, \qquad \eta_{i,\pm} \mapsto \eta_{i,\pm}/\sqrt{c}, \quad \text{for } c \in \{0.1, 0.25, 0.5, 2, 4, 8\}.$$

Remaining starts cycle through four high-jump/high-diffusion scenarios and add bounded random perturbations to scale, probability, and correlation parameters. The application calibration uses 30 starts (the library default is 20) and returns the bounded L-BFGS-B minimizer with the lowest penalized objective.

8. Empirical showcase output

The application uses four-hour observations from 2026-02-26 to 2026-05-27 and calibrates the residual shape in the market-data order WETH/WBTC. The shape-only optimization converged with zero-based best-start index 24 among 30 starts, using 234 Fourier points, with penalized objective 0.00380646. Reordering the fitted law after comparing the two role-adjusted candidate assignments gives the shape parameters in Table 2.

Parameter	WBTC	WETH
σ_i	0.256377	0.313671
λ_i	321.430466	609.872146
p_i	0.493134	0.413228
$\eta_{i,+}$	0.010557	0.011322
$\eta_{i,-}$	0.004586	0.005592

Table 2: Penalized shape-only ECF estimates, reordered to the selected long/short asset order. The fitted Brownian correlation is $\rho = 0.841703$.

Result-generation script: `jobs/calibrate_eth_btc.py`

Table 4 records the separate drift construction. The endpoint EWM signals are 0.095089 for WBTC and -0.555894 for WETH. Table 3 first enumerates both ordered assignments after applying candidate-specific supply and borrowing rates in expected-log-return space. Long WBTC/short WETH has the maximal positive spread, 0.623184, whereas the reverse assignment has a negative spread. Only after this selection are the fitted shape parameters used in (22) to derive μ_i^X and μ_i, as reported in Table 4.

Candidate long L	Candidate short S	$g_L(L,S)$	$g_S(L,S)$	$g_L(L,S) - g_S(L,S)$
WBTC	WETH	0.095145	-0.528039	0.623184
WETH	WBTC	-0.533993	0.098569	-0.632562

Table 3: Enumeration of the two role-adjusted candidate assignments. The rule selects the candidate with the largest positive expected-log-return spread, hence long WBTC and short WETH.

Result-generation script: `jobs/calibrate_eth_btc.py`

Annualized component	Long WBTC	Short WETH
Endpoint EWM log-return mean d_i^{EWM}	0.095089	−0.555894
Role-specific supply/borrowing rate	0.0000556	0.027855
Expected log-return mean g_i	0.095145	−0.528039
Mean jump rate $\lambda_i \bar{J}_i$	0.926182	0.852120
Log-SDE coefficient μ_i^X	−0.831037	−1.380159
Residual price-growth correction μ_i^{res}	0.054131	0.093001
Expected-price growth exponent μ_i	0.149275	−0.435038

Table 4: Empirical drift construction for the selected long–short orientation. The first three rows construct the expected log-return mean. The remaining rows show how the fitted residual shape maps that mean into the two drift conventions; neither μ_i^X nor μ_i is a free ECF drift estimate.

Result-generation script: `jobs/calibrate_eth_btc.py`

9. Identifiability and model-comparison protocol

The penalized shape criterion $\mathcal{Q}_{N_{\text{res}}}(\vartheta)$ is non-convex. Jump-diffusion models possess approximate ridges along directions that preserve the compound-Poisson variance contribution $\lambda_i \mathbb{E}[J_i^2] = 2\lambda_i(p_i \eta_{i,+}^2 + (1-p_i)\eta_{i,-}^2)$ (Ramezani and Zeng, 2007). The diffusion volatilities σ_i and the Brownian correlation ρ are generally better identified from short samples than jump-size and intensity parameters, which carry larger finite-sample uncertainty unless the observation window contains many jump events. Because λ_i is annualized, a one-year window contains λ_i jumps in expectation, irrespective of whether prices are sampled hourly or daily. Finer sampling can make individual jumps easier to separate from aggregated returns, but it does not multiply the underlying annual Poisson count.

For descriptive fit comparisons on a common grid, compare the unpenalized component $Q_{N_{\text{res}}}^{\text{ECF}}(\widehat{\vartheta})$ across nested residual laws (Gaussian, Merton, bivariate Kou), rather than the penalized values $\mathcal{Q}_{N_{\text{res}}}(\widehat{\vartheta})$ whose anchors differ. Such comparisons should be accompanied by liquidation-relevant diagnostics: empirical versus model Z-tail quantiles, empirical versus model barrier-crossing frequencies, and simulated versus observed drawdowns of Z. These are more directly informative for the sizing application than standard likelihood-based criteria. The present empirical illustration does not carry out a formal GBM–Merton–Kou selection exercise. Its point calibration and sizing results remain conditional on the selected Kou model and stated empirical choices; they are not evidence of model dominance or parameter-robust recommendations.

The repository's calibration package contains the implementation. The synthetic calibration job tests recovery of the eleven residual-shape parameters under the same zero-expected-log-drift restriction and produces Figure 2.

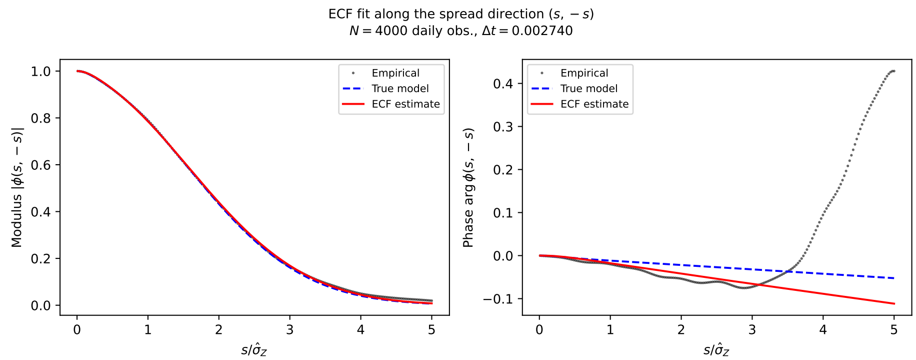

Figure 2: Shape-only ECF fit along the spread direction $(s, -s)$ for $N = 4{,}000$ synthetic daily observations from a residual law with zero expected log drift. Left: modulus $|\phi(s, -s)|$; right: phase $\arg \phi(s, -s)$. Dots: empirical CF $\widehat{\phi}_N$; dashed: true residual-law CF; solid red: shape-only ECF estimate. Agreement in this liquidation-relevant direction is a shape diagnostic; it does not validate the directional-drift forecast.
Result-generation script: `jobs/calibrate_kou.py`

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